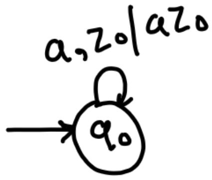
Q. Design PDA for the language $L = \{ww^R \mid w \in (a+b)^*\}$ (7 Marks)

⇒ if $w = ab$, $w^R = ba$ $\rightarrow ww^R = abba \rightarrow$ length$(ww^R) = 4 \rightarrow$ even

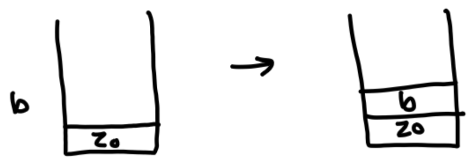
if $w = aba$, $w^R = aba$ $\rightarrow ww^R = abaaba \rightarrow l(ww^R) = 6 \rightarrow$ even

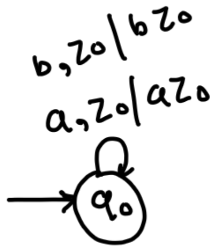
* assume that when input symbol is 'a' and stack is Z_0 (i.e. empty). then push 'a' onto the stack.

→ here, aZ_0 is the content of stack

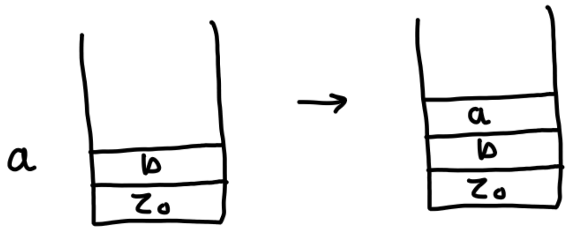


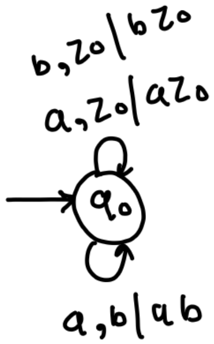
* assume that when input symbol is 'b' and stack is Z_0 (i.e. empty). then push 'b' onto the stack.





* assume that stack's top most symbol is 'b', and input symbol is 'a'.





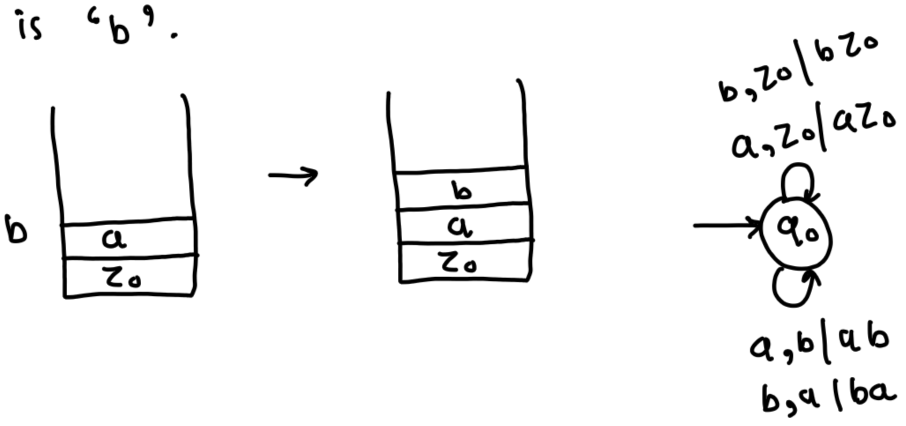
→ here, as middle part of the string is not reached, we can push 'a' onto the stack.

What is middle part of the string ?

$ww^R = a\underline{bb}a$ $\qquad$ $ww^R = ab\underline{aa}ba$

→ when last symbol of first string (w) and first symbol of second string (w^R) is same

* assume that stack's top most symbol is 'a', and input symbol is 'b'.



* assume that stack's top most symbol is 'a', and input symbol is 'a'.

→ two possibilities :

(1) Center may reached. → perform pop operation and change the state. ex. abaab

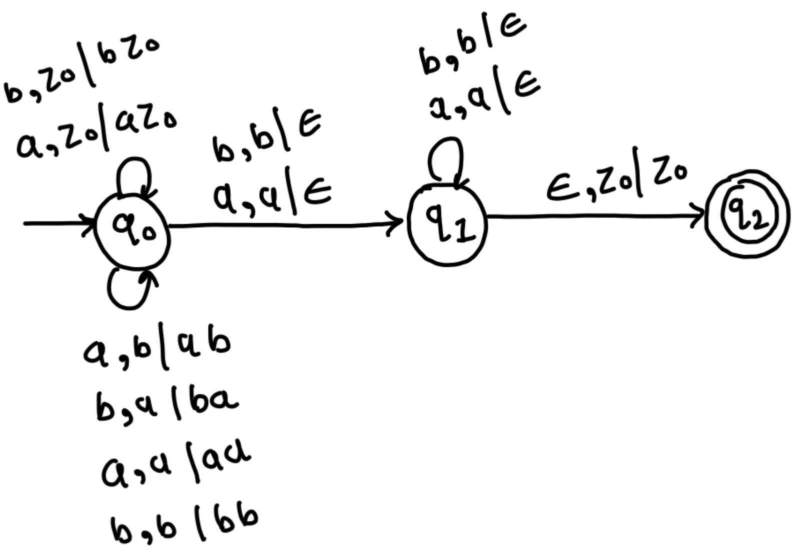
(2) center may not reached. → push input symbol on stack. ex. aaaaaa

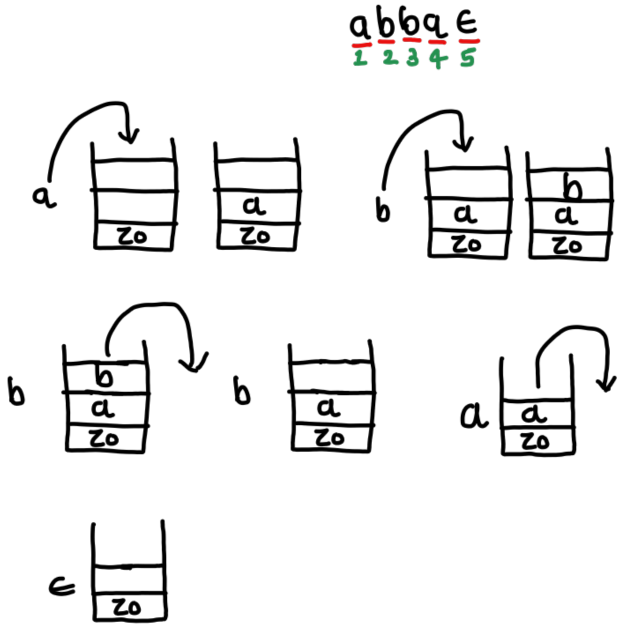
∴ because of two situations, instead of PDA, we need to use NPDA.

NPDA : If we apply input symbol on stack, there may be possibilities that we may get two states.

→ on applying same input symbol on current state, we are going in two different states.

PDA for ww^R





Transition for the string abba

$(q_0, abba, z_0)$

$\vdash (q_0, bba, az_0)$ Push 'a'

$\vdash (q_0, ba, baz_0)$ Push 'b'

$\vdash (q_1, a, az_0)$ Pop 'b'

$\vdash (q_1, \epsilon, z_0)$ Pop 'a'

$\vdash (q_2, \epsilon, z_0)$ ϵ move

Transition for the string aaaa

$(q_0, aaaa, z_0)$

$\vdash (q_0, aaa, az_0)$ Push 'a'

$\vdash (q_0, aa, aaz_0)$ Push 'a'

$\vdash (q_1, a, az_0)$ Pop 'a'

$\vdash (q_1, \epsilon, z_0)$ Pop 'a'

$\vdash (q_2, \epsilon, z_0)$ ϵ move

4 Marks Questions.

Q. Design PDA for $L = \{xcy\} \rightarrow |x| = |y|$

Q. Design PDA for $L = \{xcx^R\}$ or $L = \{wcw^R\}$

Q. Design PDA for $L = \{a^n b^n \mid n \geq 0\}$

Q. Design PDA for $L = \{a^i b^j c^k \mid j = i+k\}$

7 Marks Questions.

Q. Design PDA to accept string with more a's than b's. Also trace out the same on input string 'abbabaa'

Q. Design & Draw PDA to accept "Balance string of brackets"

Watch Sudhakar Atchala videoes on YouTube